CO326 Edge AI + Industrial IoT Mini Project (2 Weeks)

□ Objective

Design and implement a **complete Edge AI-based IoT system** that:

- Acquires sensor data (real or simulated)
- Processes data locally (edge computing)
- Applies simple AI / anomaly detection
- Communicates using MQTT
- Visualizes results in a dashboard
- Maintains code and documentation using GitHub

□ Tools & Technologies

You will use:

- Docker & Docker Compose
- Python (MQTT + AI processing)
- MQTT Broker (central cloud server)
- Node-RED (dashboard & flows)
- GitHub (code management & submission)

Optional:

- ESP32 / Raspberry Pi
- TensorFlow Lite

□ System Architecture

Sensor/Data → Python (Edge AI) → MQTT (Cloud) → Node-RED → Dashboard

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□ Development Setup

Students will:

- Run Python + AI locally
- Connect to **central MQTT broker (provided by instructor)**
- Push all code to GitHub

□ Lab Tasks

□ Task 1 — Setup Environment (Day 1–2)

Steps:

1. Install Docker
2. Clone your project repository from GitHub
3. Run provided environment:

```
docker-compose up --build
```

4. Verify:

- MQTT connection works
- Node-RED accessible

□ Task 2 — GitHub Repository Setup (Day 1–2)

Steps:

1. Create a repository on GitHub:

```
edge-ai-groupXX
```

2. Use the provided template repository
3. Maintain structure:

```
project-root/
├── python/
├── node-red/
├── docs/
```

└── README.md

✅Requirements:

- Commit code regularly
- Clear folder structure
- Proper README

□ Task 3 — Data Acquisition (Day 3–4)

Steps:

- Run `mqtt_publisher.py`
- Modify:
  - Sensor data OR dataset

MQTT Topic Format:

sensors/<group-id>/<project>/data

□ Task 4 — MQTT Communication (Day 5–6)

Steps:

- Connect to instructor's MQTT server
- Publish sensor data
- Verify using Node-RED

□ Task 5 — Dashboard Visualization (Day 7)

Using Node-RED:

- Create:
  - Chart
  - Gauge

    - Text display

□ Task 6 — Edge AI / Intelligence (Day 8–9)

Implement:

- Threshold-based detection OR
- Simple ML model

Example:

Temperature > 35 → Alert

□ Task 7 — Alert Integration (Day 10–11)

Publish alerts:

alerts/<group-id>/<project>/status

Display in Node-RED dashboard

□ Task 8 — Testing & Debugging (Day 12–13)

Check:

- Data flow
- AI logic
- MQTT communication
- Dashboard

□ Task 9 — Final Demo (Day 14)

Demonstrate:

- Live system

- AI results
- Alerts

□ Deliverables

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1. GitHub Repository (MANDATORY)

Submit repository link:

https://github.com/<username>/edge-ai-groupXX

Repository must include:

```
├── python/          (scripts)
├── node-red/        (flows.json)
├── docs/            (report)
├── README.md
```

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2. README.md (MANDATORY)

Must include:

Project Title

Group Members

Project Description

System Architecture

How to Run

MQTT Topics Used

Results (screenshots)

Challenges

Future Improvements

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3. Report (5–10 pages)

Upload to GitHub (`docs/` folder)

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4. Live Demo

□ Evaluation Scheme

Criteria	Marks
System functionality	25
Edge AI implementation	20
MQTT communication	15
Dashboard design	10
GitHub usage & code quality	15
Documentation (README + report)	10
Innovation	5

□ GitHub Best Practices

- Commit frequently (not one final upload)
- Use meaningful commit messages

- Maintain clean structure
- Avoid uploading unnecessary files

□ Important Rules

✕No ZIP file submissions
✕No code outside GitHub
✕No last-day uploads

✓ Continuous development required

□ Suggested Topics

- See attached 35 projects

□ Outcome

By completing this lab:

- You will build a **real Edge AI system**
- Learn **industrial IoT architecture**
- Gain experience with **GitHub workflows**
- Contribute to a **larger integrated system platform**